



United International University (UIU)  
Department of Computer Science and Engineering  
CSE 1325: Digital Logic Design, Midterm Fall 2025  
Total Marks: 30, Duration: 1 hour 30 minutes

**[Any examinee found adopting unfair means including copy from another examinee will be expelled from the trimester/program as per UIU disciplinary rules.]**

**Answer All Questions**

1. a) The excess three representation of eight decimal digits are as follows. [3]

$X_1 = 1011, X_2 = 1001, X_3 = 1000, X_4 = 0110, X_5 = 1000, X_6 = 0110, X_7 = 0100, X_8 = 0111.$

Find out the BCD sum of  $X_8X_5X_3X_2$  and  $X_1X_4X_6X_7$ .

- b) Find out the value of radix  $r$  and  $x$  from the following equations. [3]

i)  $x = r + 3$

ii)  $30_r \div 3_x = 66_x \div 10_{10}$

2. a) Simplify the following Boolean Expression (using **boolean algebra**) to an expression containing minimum number of literals. You have to write the identity/formula you use in every step. [3]

$$F(A, B, C, D) = \overline{\{(A\bar{B} + \bar{C}D) + D\}}\{(A + \bar{C})\bar{D}\}$$

- b) Convert the following expression into both canonical SoP (SOM) and canonical PoS (POM) forms using Boolean algebra. [3]

$$F(A, B, C, D) = A(B + \bar{C}) + \bar{A}C + A\bar{C}D$$

3. Simplify the following Boolean function

$$F(A, B, C, D) = \prod_M(1, 4, 9, 12, 14, 15)$$

by finding:

- (i) all prime implicants, [2]

- (ii) all essential prime implicants and [2]

- (iii) applying the selection rule. [2]

4. For the following Boolean function

$$F(A, B, C, D) = A'C'D' + A'CD + A'BC + BC'D + AB'C' + ABC'D$$

Find the following using K-Map [1]

- I. Simplified sum-of-products (SOP) and [2]
- II. Simplified product-of-sums (POS) form. [2]
- III. Between minimized SOP and POS, which form will you choose to implement the function and why? [1]

5. Erica is redecorating her apartment and being the independent girl she is, she decided to build a smart checker that tells her whether a chosen combination of furniture fits her budget. She has the following four pieces of furniture in mind.

Almirah (A) - Tk. 25000 ,  
Bookshelf (B) - Tk. 18000,  
Chair (C) - Tk. 5000 and  
Desk (D) - Tk. 12000.

Erica's budget is Tk. 30000. In the smart checker, each item is represented by a binary input (1 = selected, 0 = not selected).

Design a combinational circuit that outputs 1 if the total cost of the selected items is within her budget ( $\leq$  Tk. 30000). It should output 0 if the total cost exceeds her budget. The output is don't care (X) if no furniture is selected at all.

Example:

- 0001  $\rightarrow$  1 (Desk only  $\rightarrow 12,000 \leq 30,000$ ; can buy)
- 1100  $\rightarrow$  0 (Almirah and Bookshelf  $\rightarrow 43,000 > 30,000$ ; can't buy)
- 0000  $\rightarrow$  X (No furniture is selected; don't care)

You have to

- i) Draw the truth table. [2]
- ii) Find the simplified expression for the output bit in Sum-of-Product (SOP) form using K-map. [3]
- iii) Draw the circuit diagram using basic gates. [1]